QUANTITATIVE ANALYSIS

Chart Reference Guide

Complete catalogue of charts for quantitative financial analysis. Organised by topic with descriptions, use cases, and implementation notes.

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1. Price and Returns

These are the foundation charts. Every quant project starts here. They show what happened to the price and what the return characteristics look like. Without these, nothing else makes sense.

1.1 Price History with Volume

[USED IN PORTFOLIO]

A two-panel chart. The top panel shows the closing price over time. The bottom panel shows daily trading volume as bars. This is the first chart anyone looks at. It gives the overall story of the asset in one picture.

It answers the most basic question. Did the price go up or down, and when? Volume spikes usually coincide with big news, earnings, or panic selling. High volume on a price drop is more concerning than low volume on a price drop.

> *Tip: overlay key events (earnings dates, product launches) as vertical lines. This connects the price action to real decisions.*

1.2 Returns Time Series

[USED IN PORTFOLIO]

Daily returns plotted over time. Returns are calculated as the percentage change in closing price from one day to the next. This chart strips out the trend and shows the volatility structure. You can see volatility clustering (big moves followed by big moves) and spot outlier days immediately.

This is more useful than the price chart for risk analysis. Two stocks can have the same total return but completely different return profiles. One might be smooth, the other might be full of 5% daily swings.

> *Tip: use log returns for statistical modelling (they are additive over time). Use simple returns for reporting (they match what people see in their brokerage account).*

1.3 Cumulative Returns

[USED IN PORTFOLIO]

Shows how $1 invested at the start of the period grows over time. Computed as the cumulative product of (1 + daily return). This is the chart that tells the performance story. It is common to overlay a benchmark (S&P 500, sector ETF) to show relative performance.

The shape matters as much as the endpoint. A stock that compounds steadily to +50% is very different from one that hit +80% then crashed back to +50%. The cumulative return chart shows both outcomes. The drawdown chart (1.4) makes the second pattern explicit.

1.4 Drawdown Profile

[USED IN PORTFOLIO]

Shows the percentage decline from the running peak at every point in time. A drawdown of -20% means the price is 20% below its previous highest point. The chart is always negative or zero (it touches zero at new all-time highs).

This is the most important risk chart for any investor. The maximum drawdown tells you the worst pain you would have experienced if you bought at the worst possible time. The recovery time (how long the drawdown lasts before returning to zero) tells you how patient you need to be.

A stock with high returns but a -60% drawdown is very different from one with slightly lower returns but a -15% max drawdown. Most investors cannot stomach the former, regardless of the long-term payoff.

> *Tip: annotate the max drawdown point with the date and depth. Report recovery time in trading days.*

1.5 Candlestick Chart

[NOT YET USED]

Shows four data points per day: open, high, low, close (OHLC). Green (or white) candles indicate the close was above the open. Red (or black) candles indicate the close was below. The "wicks" show the intraday high and low.

Candlestick charts are the standard in technical analysis. They show information that a simple line chart hides, particularly the intraday range and whether buyers or sellers controlled the day. Patterns like doji, hammer, and engulfing are defined by candlestick shapes.

They are most useful at shorter timeframes (daily, weekly). At monthly or yearly scale, a line chart is cleaner. For quant work, candlesticks are more of a presentation tool than an analytical one, but they look professional and clients expect them.

> *Tip: use the mplfinance library in Python. It handles candlestick rendering cleanly. Limit the date range to 6-12 months for readability.*

2. Distribution Analysis

These charts examine the shape of the return distribution. Financial returns are not normally distributed. They have fat tails, skewness, and excess kurtosis. Understanding the true distribution is essential for risk modelling, option pricing, and any statistical test.

2.1 Return Histogram with Fitted Distributions

[USED IN PORTFOLIO]

A histogram of daily returns overlaid with theoretical distribution fits (Normal, Student-t, Generalised Error Distribution, Skewed-t). The histogram shows the empirical distribution. The fitted curves show how well standard models approximate reality.

The key thing to look for is the tails. If the histogram has more extreme observations than the normal curve predicts, the returns are fat-tailed. This means risk models based on normality (like basic VaR) will underestimate the probability of large losses. The Student-t distribution typically fits better because it has heavier tails.

> *Tip: zoom in on the tails separately (left tail below -3 sigma, right tail above +3 sigma) to make the fat-tail effect visible. The centre of the distribution usually looks fine for every model. The tails are where they diverge.*

2.2 QQ Plot (Quantile-Quantile)

[USED IN PORTFOLIO]

Plots the quantiles of the actual return data against the quantiles of a theoretical distribution. If the returns follow the theoretical distribution perfectly, the points lie on a straight diagonal line. Deviations from the line show where the fit breaks down.

S-shaped deviations at the ends mean fat tails. If the lower-left points curve downward and the upper-right points curve upward, the actual distribution has heavier tails than the theoretical one. This is the classic finding for financial returns against a normal distribution.

QQ plots are more powerful than histograms for spotting distributional problems because they amplify the differences in the tails. Two distributions can look identical in a histogram but show clear separation in a QQ plot.

> *Tip: always make two QQ plots, one against the normal distribution and one against the best-fitting alternative (usually Student-t). The contrast demonstrates your model selection.*

2.3 Empirical CDF vs Theoretical CDF

[NOT YET USED]

The empirical cumulative distribution function (CDF) is a step function that shows the proportion of observations below each value. Overlay the theoretical CDF from a fitted distribution. The gap between the two curves shows where the model over or underpredicts.

This is the visual equivalent of the Kolmogorov-Smirnov test. The maximum vertical gap between the empirical and theoretical CDFs is the KS test statistic. The CDF plot makes the entire distribution comparison visible, not just the tails or the centre.

> *Tip: useful as a complement to the QQ plot. QQ plots emphasise tail behaviour, CDF plots give a more balanced view of the entire distribution.*

2.4 Box Plot / Violin Plot of Returns

[NOT YET USED]

Box plots show the median, interquartile range, and outliers. Violin plots add a kernel density estimate on each side, showing the full shape of the distribution. When comparing multiple assets or time periods, these are more compact than overlaid histograms.

Group by year or by quarter to show how the return distribution changes over time. A widening violin indicates increasing volatility. Outlier dots beyond the whiskers flag extreme events.

> *Tip: for multi-asset comparison (like peer analysis), side-by-side violins are much cleaner than overlapping histograms.*

2.5 Rolling Statistics (Mean, Std, Skew, Kurtosis)

[NOT YET USED]

A 2x2 or 4x1 panel chart showing rolling window estimates of the mean, standard deviation, skewness, and kurtosis of returns. Typical window sizes are 63 days (1 quarter), 126 days (6 months), or 252 days (1 year).

This chart answers a critical question. Are the statistical properties of the returns stable over time? If the rolling kurtosis spikes during certain periods, it means the tail risk is not constant. If the rolling mean drifts, the expected return is regime-dependent. Most risk models assume these properties are stable. This chart tests that assumption.

> *Tip: add horizontal lines for the full-sample values. The distance between the rolling estimate and the full-sample estimate shows how misleading the full-sample number can be.*

3. Volatility Modelling

Volatility is not constant. It clusters, spikes, and mean-reverts. These charts visualise different approaches to measuring and forecasting volatility. Understanding which model to use and when is a core quant skill.

3.1 GARCH Conditional Volatility

[USED IN PORTFOLIO]

The conditional volatility estimated by a GARCH model, plotted over time. GARCH (Generalised Autoregressive Conditional Heteroskedasticity) models volatility as a function of past returns and past volatility. The output is a daily volatility estimate that adapts to market conditions.

This is the industry standard for volatility modelling. The chart shows periods of high and low volatility, and critically, how quickly volatility mean-reverts after a shock. High persistence (alpha + beta close to 1) means shocks decay slowly. Low persistence means volatility snaps back quickly.

Common variants include GARCH(1,1), GJR-GARCH (which captures the leverage effect, where negative returns increase volatility more than positive returns), and EGARCH. For equities, GJR-GARCH typically fits better. For crypto, standard GARCH is often sufficient because the leverage effect is weaker.

> *Tip: overlay multiple GARCH variants (GARCH, GJR-GARCH, EGARCH) on the same chart. They usually track closely in calm markets but diverge during stress. The divergence shows which model assumption matters.*

3.2 EWMA Volatility

[USED IN PORTFOLIO]

Exponentially Weighted Moving Average volatility. A simpler alternative to GARCH with one parameter (lambda, typically 0.94 for daily data per RiskMetrics). Recent observations get more weight. It adapts faster than a simple rolling standard deviation but has no mean-reversion.

EWMA is useful as a benchmark. If GARCH produces very similar results to EWMA, the extra complexity of GARCH may not be justified. If they diverge, GARCH is capturing something EWMA misses (usually mean-reversion or the leverage effect).

3.3 Realised Volatility Comparison

[NOT YET USED]

Plots multiple volatility estimators side by side. Typical choices are the close-to-close estimator (standard deviation of returns), Parkinson (uses high-low range), Garman-Klass (uses open-high-low-close), and Yang-Zhang (the most efficient daily estimator).

Different estimators use different price information. Close-to-close ignores the intraday range. Parkinson uses only the high and low. Garman-Klass uses all four OHLC prices and is roughly 8x more efficient than close-to-close (meaning it gives a tighter estimate from the same number of observations).

Comparing estimators tells you whether the intraday range contains useful volatility information beyond what the close-to-close measure captures. If Garman-Klass is consistently higher than close-to-close, there is significant intraday movement that closing prices miss.

> *Tip: a good chart uses a rolling window (20 or 30 days) and plots all estimators on the same axes. Annotate which estimator is most appropriate for your use case.*

3.4 Volatility Cone

[NOT YET USED]

Shows the distribution of realised volatility at different lookback periods. For each lookback (10, 20, 30, 60, 90, 120, 252 days), compute the rolling realised volatility across the entire history, then plot the percentiles (10th, 25th, 50th, 75th, 90th) as lines. The current realised volatility is overlaid as a point at each lookback.

The cone shape emerges because short-lookback volatility is more variable (the 10-day vol swings widely) while long-lookback volatility is more stable (the 252-day vol is smoother). The cone tells you whether current volatility is high or low relative to history, at every timescale.

This is widely used by options traders. If the current 30-day realised vol is at the 90th percentile, implied volatility should be priced accordingly. If it is at the 10th percentile, the market may be underpricing risk.

> *Tip: this chart impresses interviewers. It shows you understand volatility across multiple timescales, not just the headline number.*

3.5 Implied vs Realised Volatility

[NOT YET USED]

Plots implied volatility (from options prices) against realised volatility (from actual returns) over time. The gap between them is the volatility risk premium. Implied volatility is almost always higher than realised, because options sellers demand compensation for the risk of being wrong.

When the gap widens, options are expensive relative to actual risk. When it narrows or inverts, options are cheap. This is the foundation of volatility trading strategies. The chart also shows how well the options market predicts future volatility. If implied vol consistently overshoots, there is a systematic premium to harvest.

> *Tip: requires options data (VIX for the S&P 500, or individual stock IV from options chains). The VIX vs S&P 500 realised vol comparison is the most accessible version.*

4. Risk Metrics

Value at Risk (VaR), Expected Shortfall (CVaR), and related metrics. These charts translate statistical properties into dollar-denominated risk numbers that portfolio managers and risk committees understand.

4.1 VaR Comparison (Multiple Methods)

[USED IN PORTFOLIO]

Shows the daily return series with VaR estimates from multiple methods overlaid as horizontal or time-varying lines. Methods include Historical VaR, Parametric Normal, Parametric Student-t, GARCH-based, and Monte Carlo. The chart shows how different assumptions produce different risk estimates.

The key insight is that Normal VaR consistently underestimates tail risk. The gap between Normal and Student-t VaR quantifies how much tail risk the normal assumption misses. GARCH-based VaR adapts over time (higher in volatile periods, lower in calm ones), which is more realistic than a static estimate.

> *Tip: use a time-varying version where the VaR lines move with conditional volatility. Static VaR lines are a simplification.*

4.2 VaR Backtest (Violations Chart)

[USED IN PORTFOLIO]

Plots the return series with the VaR threshold as a line. Days where the actual loss exceeds the VaR estimate are marked with coloured dots (violations). The violation rate should match the confidence level. At 99% VaR, roughly 1% of days should be violations.

This is the definitive test of a VaR model. Too many violations means the model underestimates risk. Too few means it is overly conservative (which is not necessarily bad, but it ties up more capital than needed). The Kupiec test formalises this by testing whether the violation rate is statistically consistent with the target.

Violations should also be independent (not clustered). If violations cluster, it means the model fails during exactly the periods when it matters most. The Christoffersen test checks for this.

> *Tip: mark violation clusters with a different colour or annotation. Clustered violations are more damaging than scattered ones.*

4.3 Expected Shortfall (CVaR) Comparison

[USED IN PORTFOLIO]

Similar to the VaR comparison, but for CVaR (Conditional Value at Risk, also called Expected Shortfall). CVaR measures the average loss on days that exceed the VaR threshold. It answers: "When things go bad, how bad do they get on average?"

CVaR is a more conservative and coherent risk measure than VaR. VaR tells you the boundary of the "bad zone." CVaR tells you the average depth of the bad zone. Two portfolios can have the same VaR but very different CVaR if one has a longer left tail.

> *Tip: report both VaR and CVaR. VaR alone can be misleading for assets with asymmetric tails.*

4.4 Loss Exceedance Curve

[NOT YET USED]

Plots the probability that a loss exceeds a given threshold, for all thresholds. The x-axis is the loss size (in percent or dollars). The y-axis is the probability of exceeding that loss. This is the survival function of the loss distribution.

It provides more information than a single VaR number. Instead of saying "the 99% VaR is 4.5%," it shows the entire tail. You can read off the VaR at any confidence level directly from the curve. Overlaying the empirical curve with a fitted model (GPD, Student-t) shows where the model diverges from reality.

> *Tip: use a log scale on the y-axis. This stretches the tail and makes it easier to see differences at extreme probabilities (99.5%, 99.9%).*

4.5 Stress Test Scenario Chart

[NOT YET USED]

A bar chart or waterfall showing portfolio or asset performance under named stress scenarios. Scenarios might include: 2008 financial crisis, COVID-19 crash, 2022 rate shock, hypothetical 50% revenue decline, or sector-specific events.

Stress tests go beyond statistical risk measures. They ask "what happens if history repeats?" VaR and CVaR are backward-looking statistical summaries. Stress tests are forward-looking and scenario-specific. They are especially useful for communicating risk to non-technical stakeholders, because "the portfolio would have lost 28% in a 2008-style event" is more tangible than "the 99% VaR is 4.5%."

> *Tip: include both historical scenarios (replay actual bad periods) and hypothetical scenarios (imagine plausible future shocks). The historical ones are defensible. The hypothetical ones test imagination.*

5. Tail Risk and Extreme Value Theory

The tails of the distribution are where the money is lost. Standard models underestimate tail risk because they assume returns are normally distributed or fit a single distribution to the entire range. Extreme Value Theory (EVT) focuses specifically on the tail.

5.1 Mean Excess Plot

[NOT YET USED]

For a given threshold u, the mean excess is the average amount by which observations exceed u, conditional on exceeding it. Plot this for all values of u. If the mean excess plot is approximately linear and increasing, it suggests a fat-tailed distribution (Pareto-type). If it levels off, the tail is lighter.

This chart is used to select the threshold for the Peaks Over Threshold (POT) method in EVT. The threshold should be chosen in the region where the mean excess plot is approximately linear. Below that, you are contaminating the tail sample with non-extreme observations. Above it, you have too few observations for a reliable fit.

> *Tip: combine with the Hill plot (5.2) for threshold selection. They should agree. If they do not, the tail estimation is fragile.*

5.2 Hill Plot

[USED IN PORTFOLIO]

Plots the Hill estimator of the tail index as a function of the number of order statistics used. The Hill estimator gives the shape parameter of the tail (related to the Generalised Pareto shape parameter). The plot should stabilise (flatten out) for a range of order statistics. The value where it stabilises is the tail index estimate.

If the Hill plot does not stabilise, the tail index is hard to estimate reliably. This is common with financial data, where the number of extreme observations is small. The instability itself is an important finding. It means any specific tail risk number should be reported with wide confidence intervals.

5.3 GPD Fit to Tail

[USED IN PORTFOLIO]

Shows the empirical distribution of exceedances above a threshold overlaid with the fitted Generalised Pareto Distribution (GPD). This is the core EVT chart. It visually confirms (or refutes) that the GPD is a reasonable model for the tail.

The shape parameter (xi) is the key output. If xi > 0, the tail is heavier than exponential (Pareto-type, which is typical for financial returns). If xi = 0, the tail is exponential. If xi < 0, the tail has a finite upper bound. For most financial assets, xi is slightly positive (0.1 to 0.4).

5.4 EVT Threshold Sensitivity

[USED IN PORTFOLIO]

Fits the GPD at multiple threshold values and plots how the shape parameter (xi) and scale parameter change. If the parameters are stable across a range of thresholds, the EVT model is robust. If they swing wildly, the tail estimation is fragile and the specific threshold choice dominates the result.

This chart is essential for honest EVT analysis. It prevents the common mistake of reporting a single GPD fit as if it were precise, when in fact the result depends heavily on an arbitrary threshold choice.

Tip: plot xi with bootstrap confidence intervals at each threshold. If the CIs overlap across thresholds, the model is stable. If they do not, report the range.

6. Monte Carlo Simulation

Monte Carlo methods generate thousands of possible future paths by sampling from a model. They are used for pricing, risk estimation, and scenario analysis. The power of Monte Carlo is that it handles non-linearity and complex distributions naturally.

6.1 Price Fan Chart (Simulation Paths)

[USED IN PORTFOLIO]

Shows simulated future price paths radiating from the current price. A handful of individual paths are drawn as thin lines to show the range of possibilities. The median path and percentile bands (5th/25th/75th/95th) are overlaid as thicker lines or shaded regions.

The "fan" shape makes uncertainty tangible. The further into the future, the wider the fan. This communicates that uncertainty grows with time, which is obvious in theory but powerful when visualised. The percentile bands answer: "What is the range of likely outcomes at 30, 60, 90 days?"

> *Tip: use GBM (Geometric Brownian Motion) as the baseline model. For assets with jumps (crypto, volatile stocks), overlay Merton jump-diffusion paths to show how jumps widen the fan.*

6.2 Terminal Value Distribution

[USED IN PORTFOLIO]

A histogram of the simulated values at the end of the projection period. This is the Monte Carlo version of a fair value distribution. For DCF models, it shows the distribution of implied fair values. For price simulations, it shows where the price might end up.

The shape of this distribution is informative. If it is right-skewed (long right tail), there is a small probability of very large gains. If it is left-skewed, there is a small probability of very large losses. The median, mean, and mode are often different, which matters for decision-making.

> *Tip: mark the current price on the histogram. The proportion of simulated values above the current price is the probability the asset is undervalued on that model.*

6.3 Sensitivity Tornado Chart

[USED IN PORTFOLIO]

A horizontal bar chart showing how sensitive the output (e.g., fair value) is to each input parameter. Each bar shows the range of outputs when one parameter varies while others are held at their base case. The longest bar is the most influential parameter.

This answers: "Which assumptions matter most?" If the WACC bar is three times longer than the growth rate bar, then getting the WACC right is three times more important than getting growth right. It focuses the analysis on the parameters that actually drive the conclusion.

> *Tip: an alternative is the correlation-based tornado, where each bar shows the correlation between the input and the output across Monte Carlo simulations. This captures non-linear effects that the one-at-a-time approach misses.*

6.4 Sensitivity Heat Map

[USED IN PORTFOLIO]

A matrix where two input parameters vary along the axes and the output value is shown as colour intensity in each cell. For a DCF, the typical axes are WACC and revenue growth, with fair value as the colour.

This shows two-way interactions that a tornado chart cannot. It answers: "What happens if both WACC increases AND growth slows?" The corners of the heat map represent the extreme scenarios.

7. Technical and Momentum Analysis

Technical indicators are mathematical transformations of price and volume data. They are used to identify trends, momentum, overbought/oversold conditions, and potential reversal points. In quant work, they serve as features for statistical models as much as standalone signals.

7.1 Moving Average Crossover

[USED IN PORTFOLIO]

Price with SMA 50 and SMA 200 overlaid. The "Golden Cross" (SMA 50 crosses above SMA 200) is a bullish signal. The "Death Cross" (SMA 50 crosses below SMA 200) is bearish. These crossovers are the simplest trend-following signals.

Golden and death crosses are lagging indicators. By the time they trigger, a significant portion of the move has already happened. But they are reliable in trending markets. The strategy works poorly in choppy, range-bound markets where false signals accumulate.

> *Tip: mark crossover points on the chart. Show historical performance after each signal to demonstrate whether crossovers had predictive power for this specific asset.*

7.2 RSI (Relative Strength Index)

[USED IN PORTFOLIO]

A bounded oscillator (0-100) that measures the speed and magnitude of recent price changes. Values above 70 indicate overbought conditions. Below 30 indicates oversold. The standard lookback period is 14 days.

RSI works best in range-bound markets. In strong trends, an asset can remain overbought or oversold for extended periods. Using RSI as a reversal signal in a trending market leads to systematic losses. It is most useful as a confirmation tool rather than a standalone signal.

Divergence between RSI and price is the more sophisticated use. If the price makes a new high but RSI does not, the momentum is weakening. This is called "bearish divergence" and often precedes a correction.

7.3 MACD (Moving Average Convergence Divergence)

[USED IN PORTFOLIO]

Three components: the MACD line (12-period EMA minus 26-period EMA), the Signal line (9-period EMA of the MACD), and the Histogram (difference between MACD and Signal). Bullish when MACD crosses above Signal. Bearish when it crosses below.

MACD captures momentum and trend in one indicator. The histogram is especially useful. A shrinking histogram (even if still positive) indicates momentum is fading. This can signal a reversal before the crossover happens.

> *Tip: in the chart, plot the histogram as coloured bars (green positive, red negative) and the MACD and Signal as lines. The visual contrast makes the momentum shifts obvious.*

7.4 Bollinger Bands

[USED IN PORTFOLIO]

Three lines: the middle band (20-day SMA), the upper band (SMA + 2 standard deviations), and the lower band (SMA - 2 standard deviations). Price touching the upper band suggests the asset is expensive relative to recent history. Touching the lower band suggests it is cheap.

Band width is informative. When the bands are narrow (a "squeeze"), volatility is low and a breakout may be coming. When they are wide, volatility is high and may be about to contract. The squeeze is often used as a precursor to a large move in either direction.

The %B indicator (percentage of distance between lower and upper band) reduces Bollinger Bands to a single number between 0 and 1. Values above 1 or below 0 mean the price has broken outside the bands.

7.5 Multi-Period Momentum Bar Chart

[USED IN PORTFOLIO]

A bar chart showing price returns over multiple lookback periods (1 week, 1 month, 3 months, 6 months, 1 year). Green for positive, red for negative. This gives the momentum picture at a glance.

The pattern matters more than any single bar. If all timeframes are positive, the trend is strong. If long-term is positive but short-term is negative, momentum is fading. If short-term is positive but long-term is negative, a bounce might be forming within a downtrend.

7.6 Relative Strength vs Benchmark

[USED IN PORTFOLIO]

The ratio of cumulative returns of the asset to cumulative returns of a benchmark (typically S&P 500). When the line rises, the asset is outperforming the benchmark. When it falls, it is underperforming.

This strips out market-wide moves and isolates stock-specific performance. An asset can fall in price but still show rising relative strength if the market fell more. For portfolio managers, relative performance is often more relevant than absolute performance.

7.7 Volume Profile

[NOT YET USED]

A horizontal histogram plotted along the y-axis (price axis) showing the total volume traded at each price level. High-volume price levels are areas of strong support or resistance because many shares changed hands there.

The "Point of Control" is the price level with the highest traded volume. It represents the price where the most agreement between buyers and sellers occurred. Prices tend to be attracted to the POC. Gaps in the volume profile (low-volume zones) can act as areas where price moves quickly through.

> *Tip: requires intraday or tick data for precision. Daily OHLCV data gives a rough approximation by distributing volume across the day is range.*

7.8 Hurst Exponent Visualisation

[USED IN PORTFOLIO]

A chart showing the R/S (rescaled range) analysis used to compute the Hurst exponent. Plot log(R/S) against log(lag). The slope of the fitted line is the Hurst exponent. Values above 0.5 indicate trending behaviour. Below 0.5 indicates mean-reversion. Equal to 0.5 is a random walk.

This can also be shown as a rolling Hurst exponent over time, which reveals whether the asset switches between trending and mean-reverting regimes. A stock might be trending for months, then shift to mean-reverting. Knowing which regime you are in changes the optimal strategy.

8. Event Studies

Event studies measure the impact of specific events (earnings, announcements, policy changes) on stock returns. They isolate the abnormal return caused by the event from the normal return the market would have predicted.

8.1 CAR Plot (Cumulative Abnormal Return)

[USED IN PORTFOLIO]

For a single event, plot the cumulative abnormal returns from before the event to after it. The x-axis is the number of days relative to the event (day 0 is the announcement). The y-axis is the cumulative abnormal return.

A sharp jump on day 0 means the market reacted immediately. A gradual drift before day 0 suggests information leakage (someone traded on the news before it was public). A drift after day 0 suggests the market is still digesting the information (post-earnings drift is a well-documented phenomenon).

> *Tip: add confidence bands (based on the estimation window residuals) to show whether the CAR is statistically significant at each point in the window.*

8.2 CAR Grid (Multiple Events)

[USED IN PORTFOLIO]

A grid of small CAR plots, one for each event studied. This allows visual comparison across multiple events. You can immediately see which events had the largest impact, whether the impact was positive or negative, and whether the market reaction was immediate or gradual.

This is more informative than a table of numbers. The shape of each CAR curve tells a story that a single number (total CAR) cannot. Two events with the same total CAR can have very different dynamics.

8.3 Event Timeline Overlay

[NOT YET USED]

The full price chart with vertical lines or markers at each studied event. Positive events in green, negative in red, neutral in grey. This puts the event study results in the context of the overall price history.

It answers: "How much of the price movement was driven by events versus gradual drift?" If most of the action happens around event dates, the stock is event-driven. If the price moves smoothly between events, fundamentals and momentum dominate.

9. Valuation

Valuation charts translate financial models into visual conclusions. They show what a company is worth under different assumptions and how the current price compares to that estimate.

9.1 DCF Fair Value Distribution

[USED IN PORTFOLIO]

A histogram of fair value estimates from a Monte Carlo DCF model. The current market price is marked as a vertical line. The proportion of simulations above the market price is the probability the stock is undervalued.

This is more useful than a single DCF fair value because it shows the full range of uncertainty. A point estimate of $150 means little without knowing whether the 90% CI is [$140, $160] or [$50, $400]. The histogram shows exactly this.

9.2 Peer Comparison Bar Charts

[USED IN PORTFOLIO]

Horizontal bar charts comparing the subject company to peers on key valuation metrics (P/E, EV/EBITDA, P/B, Price/Sales). The subject company is highlighted in a different colour. The peer average is shown as a vertical line.

Z-scores quantify how many standard deviations the subject is from the peer mean. A z-score above 2 in any metric is extreme. A company trading at a premium on every metric needs a strong qualitative justification.

9.3 Football Field Chart (Valuation Range)

[NOT YET USED]

Named after its shape. Shows the range of fair values from multiple valuation methods as horizontal bars. Methods might include DCF (low, mid, high), comparable company analysis, precedent transactions, 52-week range, and analyst targets.

The current price is marked as a vertical line. If the current price falls within most of the bars, the market is pricing the company reasonably. If it sits above all of them, the stock appears overvalued by every method. If below all, undervalued.

This is a standard chart in investment banking pitch decks and equity research reports. It communicates the valuation conclusion in a single image.

> *Tip: order the methods from most conservative at the top to most optimistic at the bottom. This creates a visual narrative.*

9.4 Historical Multiples Chart

[NOT YET USED]

Plots a valuation multiple (P/E, EV/EBITDA, P/B) over time with horizontal lines for the historical mean, plus and minus one standard deviation. Shows whether the stock is cheap or expensive relative to its own history.

This is distinct from peer comparison (which asks "is it cheap relative to similar companies?"). Historical multiples ask "is it cheap relative to itself?" A stock can be expensive vs peers but cheap vs its own history if the industry has re-rated.

> *Tip: requires historical earnings or EBITDA data. Trailing twelve-month earnings work for P/E. Use quarterly financial statements to build the time series.*

10. Statistical Robustness

These charts test whether your results are stable and reliable. They separate genuine findings from statistical noise. Any quant report without robustness checks is incomplete.

10.1 Bootstrap Distribution

[USED IN PORTFOLIO]

Histograms of key parameters (Sharpe ratio, VaR, beta, GARCH persistence) estimated across 1,000+ bootstrap resamples. The point estimate is marked with a vertical line. The 95% confidence interval bounds are marked with dashed lines.

Bootstrap CIs answer: "How much would this estimate change if we had a different sample of the same size?" If the CI is tight, the estimate is stable. If it is wide (or includes zero for the Sharpe ratio), the finding is uncertain. Reporting point estimates without CIs is intellectually dishonest in any statistical analysis.

> *Tip: use block bootstrap for time series data. Naive bootstrap destroys the autocorrelation structure. Block sizes of 10-30 trading days are typical.*

10.2 Parameter Stability (Rolling Windows)

[USED IN PORTFOLIO]

Plots model parameters estimated on rolling windows of fixed length. For GARCH, track alpha, beta, and persistence. For beta regression, track the rolling beta. For EVT, track the GPD shape parameter. The x-axis is time (the end date of each window).

If parameters drift substantially, the full-sample estimate is an average of different regimes. The specific value you report depends on when you happen to estimate it. This is a fundamental problem for any model that assumes stationarity.

Wide swings in rolling parameters suggest regime changes. A GARCH persistence that occasionally exceeds 1.0 in rolling windows means the model is occasionally non-stationary, even if the full-sample estimate is below 1.

10.3 CUSUM Structural Break

[USED IN PORTFOLIO]

The cumulative sum of deviations of returns from their mean, normalised by the standard deviation. The CUSUM statistic should stay within critical value bands (typically +/- 1.36 at 5%) if the process is stationary. A breach indicates a structural break.

The point of maximum deviation suggests when the break occurred. This can be a change in mean return, volatility regime, or both. CUSUM is a simple but effective test for non-stationarity.

10.4 Residual Diagnostics Panel

[NOT YET USED]

A 2x2 panel showing: (1) histogram of standardised residuals vs normal, (2) QQ plot of residuals, (3) ACF of residuals, (4) ACF of squared residuals. This is the standard model validation chart for any GARCH or regression model.

Residuals should be roughly normal (or match the assumed distribution). ACF of residuals should show no significant autocorrelation (meaning the mean model captured the serial dependence). ACF of squared residuals should show no significant autocorrelation (meaning the volatility model captured the ARCH effects). Failures indicate model misspecification.

Tip: always include this after fitting a GARCH model. It is the first thing a reviewer checks.

10.5 OOS Forecast vs Realised

[USED IN PORTFOLIO]

Plots out-of-sample forecasts (volatility or VaR) against realised values on the test set. A 45-degree line represents perfect forecasts. Points clustered along the line indicate good forecasting ability. Systematic deviations indicate bias.

The Mincer-Zarnowitz regression formalises this. Regress realised values on forecasts. The intercept should be zero and the slope should be one. Deviations from (0, 1) indicate bias or inefficiency. The R-squared shows what fraction of realised variation the forecast explains.

11. Portfolio and Multi-Asset

These charts only apply when analysing multiple assets together. They deal with correlations, diversification, and portfolio construction.

11.1 Correlation Matrix Heatmap

[NOT YET USED]

A square matrix where each cell shows the correlation between two assets, colour-coded from blue (negative) through white (zero) to red (positive). The diagonal is always 1 (an asset is perfectly correlated with itself).

High correlations (above 0.8) mean two assets move together. Low or negative correlations provide diversification. During crises, correlations tend to spike toward 1 (the "correlation breakdown" problem), which means diversification fails exactly when you need it most.

> *Tip: use a rolling correlation over time to show how correlations change. A static correlation matrix hides regime-dependent correlation structure.*

11.2 Rolling Correlation

[NOT YET USED]

The correlation between two assets estimated on a rolling window (60, 90, or 120 days), plotted over time. This shows whether the relationship between assets is stable or varies with market conditions.

Rising rolling correlation during market stress is one of the most important findings in portfolio risk. It means the portfolio is less diversified than steady-state correlations suggest. DCC-GARCH (Dynamic Conditional Correlation) formalises this, but the rolling correlation chart is simpler and communicates the same story.

11.3 Efficient Frontier

[NOT YET USED]

The curve showing the maximum return achievable for each level of risk (or equivalently, the minimum risk for each return target). Portfolios on the frontier are "efficient." Portfolios below the frontier are suboptimal.

The tangent line from the risk-free rate to the frontier identifies the maximum Sharpe ratio portfolio. This is the theoretically optimal risky portfolio. Real-world constraints (no short selling, position limits, sector constraints) distort the frontier, and these should be shown as a separate constrained frontier.

> *Tip: plot individual assets as dots alongside the frontier to show the diversification benefit. The frontier always lies to the left of (better than) any individual asset.*

11.4 Beta Scatter Plot

[NOT YET USED]

A scatter plot of asset returns vs market returns with a regression line. The slope is beta. The scatter around the line shows idiosyncratic risk. A tight scatter (high R-squared) means the asset is mostly driven by market moves. A wide scatter means stock-specific factors dominate.

This is the visual representation of CAPM. Beta above 1 means the asset amplifies market moves. Below 1 means it dampens them. The intercept (alpha) shows excess return after controlling for market risk. Positive alpha means the asset outperformed what CAPM predicted.

11.5 Risk Contribution / Decomposition

[NOT YET USED]

A stacked bar or pie chart showing how much each asset contributes to total portfolio risk. Risk contribution is not the same as weight. A small position in a volatile, highly correlated asset can contribute more risk than a large position in a stable, uncorrelated one.

This chart reveals hidden risk concentrations. A portfolio that looks diversified by weight may be concentrated by risk. If one asset contributes 60% of the risk despite being 20% of the weight, the portfolio is effectively a leveraged bet on that asset.

12. Time Series Diagnostics

Before fitting any model, you need to understand the time series structure of the data. These charts diagnose autocorrelation, stationarity, and serial dependence.

12.1 ACF Plot (Autocorrelation Function)

[NOT YET USED]

Shows the correlation of the series with its own lagged values. The x-axis is the lag (1 day, 2 days, etc.). The y-axis is the correlation. Horizontal dashed lines show the 95% significance bounds. Bars exceeding these bounds indicate significant autocorrelation.

For daily stock returns, significant autocorrelation at lag 1 would indicate predictability (which efficient markets should eliminate). For squared returns, significant autocorrelation indicates volatility clustering (which GARCH models capture). Most financial returns show no significant ACF in levels but strong ACF in squared returns.

12.2 PACF Plot (Partial Autocorrelation Function)

[NOT YET USED]

Shows the direct correlation at each lag after removing the effect of intermediate lags. The PACF cuts off sharply after the true AR order. The ACF decays gradually. Together, ACF and PACF identify the order of ARMA models.

In practice, for financial returns, both ACF and PACF should be near zero at all lags (returns are approximately unpredictable). Significant PACF at lag 1 in returns would suggest an AR(1) mean model. Significant PACF in squared returns helps select GARCH order.

12.3 Ljung-Box Test Bar Chart

[NOT YET USED]

A bar chart of Ljung-Box Q statistics at multiple lags (5, 10, 15, 20) with the 5% significance threshold overlaid. Bars exceeding the threshold indicate significant serial correlation. This is a compact summary of the autocorrelation structure.

Running the Ljung-Box test on raw returns tests for predictability in the mean. Running it on squared residuals from a GARCH model tests whether the volatility model captured all the ARCH effects. The squared residuals test is the more important one for risk modelling.

13. Factor Models and Regression

Factor models decompose returns into systematic (market, size, value) and idiosyncratic components. These charts help understand what drives an asset is returns and how exposed it is to different risk factors.

13.1 Factor Exposure Bar Chart

[NOT YET USED]

A horizontal bar chart showing the regression coefficients (betas) of the asset on multiple factors. Common factor models include CAPM (one factor: market), Fama-French 3-factor (market, size, value), and Fama-French 5-factor (adds profitability and investment).

Each bar tells you how sensitive the asset is to that factor. A high market beta means the stock moves strongly with the market. A positive value (HML) loading means the stock behaves like a value stock. A negative size (SMB) loading means it behaves like a large-cap.

13.2 Rolling Beta

[USED IN PORTFOLIO]

The market beta estimated on rolling windows, plotted over time. This shows whether the systematic risk is stable or changes with market conditions. A beta that spikes during crises means the asset becomes more correlated with the market during stress.

Beta instability undermines CAPM and any valuation model that uses a single beta for the WACC. If the rolling beta ranges from 0.7 to 1.5, any specific WACC estimate is conditional on which beta you use. The rolling chart makes this uncertainty visible.

13.3 Alpha Decomposition

[NOT YET USED]

A cumulative chart showing total return decomposed into: alpha (excess return), market contribution (beta times market return), and factor contributions. The stacked areas show what drove performance.

If most of the return comes from market exposure, the asset is not providing unique value. If alpha is large and positive, the asset is generating returns beyond what its risk factors predict. This distinction matters for portfolio construction and fee justification.

14. Options and Derivatives

These charts apply when working with options or other derivative instruments. They require options market data, which is more specialised than equity price data.

14.1 Payoff Diagram

[NOT YET USED]

Shows the profit/loss of an options strategy at expiration as a function of the underlying price. Simple positions (long call, long put) have kinked lines. Complex strategies (straddles, spreads, iron condors) have distinctive shapes that make the risk-reward profile immediately clear.

The payoff diagram answers: "What is my worst case, best case, and break-even?" The shape of the curve communicates the strategy better than any verbal description. It is the universal language of options analysis.

14.2 Implied Volatility Surface

[NOT YET USED]

A 3D surface plot where the x-axis is strike price (or moneyness), the y-axis is time to expiration, and the z-axis (colour or height) is implied volatility. The surface shows how the market prices risk across different strikes and maturities.

In a Black-Scholes world, the surface would be flat (same IV everywhere). In reality, it shows a "smile" or "skew." The skew (higher IV for out-of-the-money puts) reflects the market pricing crash risk. Changes in the surface shape over time reveal shifts in risk perception.

> *Tip: this is an advanced chart that requires options chain data. Use it for equity options analysis or volatility trading projects.*

14.3 Greeks Sensitivity

[NOT YET USED]

Line charts showing how delta, gamma, theta, and vega change as the underlying price or time to expiry changes. These are the partial derivatives of the option price with respect to market variables.

Delta tells you the directional exposure. Gamma tells you how fast delta changes. Theta is the time decay (negative for long options). Vega is the sensitivity to volatility. Plotting these as the underlying moves shows where the position is most sensitive and where the hedging costs are highest.

Summary

This guide catalogues 50+ chart types across 14 categories. Not every project needs every chart. The selection depends on the analysis type.

Risk Management Projects

Focus on: Price and returns (Section 1), Distribution analysis (Section 2), Volatility modelling (Section 3), Risk metrics (Section 4), Tail risk/EVT (Section 5), Monte Carlo (Section 6), and Statistical robustness (Section 10).

Valuation and Momentum Projects

Focus on: Price and returns (Section 1), Technical indicators (Section 7), Event studies (Section 8), Valuation (Section 9), and Statistical robustness (Section 10).

Portfolio and Multi-Asset Projects

Focus on: Price and returns (Section 1), Volatility (Section 3), Risk metrics (Section 4), Portfolio analysis (Section 11), and Factor models (Section 13).

Options and Derivatives Projects

Focus on: Volatility (Section 3), Monte Carlo (Section 6), Options-specific charts (Section 14), and any relevant underlying asset charts from other sections.

The Python scripts in this folder implement the charts marked as "USED IN PORTFOLIO" and will expand to cover the remaining chart types over time.